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 Studierichting: Informatica
 Oefeningenreeks 5L nr. 1.4

$$f(x) = \frac{x^4 + 3}{6x} \text{ op } [1, 2]$$

x	1		2
$f(x)$	$\frac{2}{3}$	+	$\frac{19}{12}$
$f'(x)$	0	+	+
$f''(x)$	+	+	+
$f(x)$	$\frac{2}{3}$	$\nearrow$ $\smile$	$\frac{19}{12}$

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$$f'(x) = \frac{x^4 - 1}{2x^2}$$

$$\begin{aligned}
 S &= 2\pi \int_1^2 f(x) \sqrt{1 + (f'(x))^2} dx \\
 &= 2\pi \int_1^2 \frac{x^4 + 3}{6x} \sqrt{1 + \left(\frac{x^4 - 1}{2x^2}\right)^2} dx \\
 &= 2\pi \int_1^2 \frac{x^4 + 3}{6x} \sqrt{1 + \frac{(x^4 - 1)^2}{4x^4}} dx \\
 &= 2\pi \int_1^2 \frac{x^4 + 3}{6x} \sqrt{\frac{4x^4 + (x^4 - 1)^2}{4x^4}} dx \\
 &= 2\pi \int_1^2 \frac{x^4 + 3}{6x} \sqrt{\frac{x^8 + 2x^4 + 1}{4x^4}} dx \\
 &= 2\pi \int_1^2 \frac{x^4 + 3}{6x} \cdot \frac{x^4 + 1}{2x^2} dx \\
 &= 2\pi \int_1^2 \frac{(x^4 + 3)(x^4 + 1)}{12x^3} dx
 \end{aligned}$$

$$\begin{aligned}
&= 2\pi \int_1^2 \frac{x^8 + 4x^4 + 3}{12x^3} dx \\
&= 2\pi \int_1^2 \left(\frac{x^5}{12} + \frac{x}{3} + \frac{1}{4x^3} \right) dx \\
&= 2\pi \left[\frac{x^6}{72} + \frac{x^2}{6} - \frac{1}{8x^2} \right]_1^2 \\
&= 2\pi \left(\left(\frac{64}{72} + \frac{4}{6} - \frac{1}{32} \right) - \left(\frac{1}{72} + \frac{1}{6} - \frac{1}{8} \right) \right) \\
&= 2\pi \left(\frac{16}{18} + \frac{2}{3} - \frac{1}{32} - \frac{1}{72} - \frac{1}{6} + \frac{1}{8} \right) \\
&= 2\pi \left(\frac{47}{32} \right) \\
&= \frac{47\pi}{16}
\end{aligned}$$

References